

# altairsim — Graphics on CADzilla: the drawing demo

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2026-09-29 · 3312d78

**drawdemo** shows every drawing command of CADzilla's HD63484 ACRTC graphics processor, one screen for each command. It runs under CP/M 2.2 on an 8" floppy.

```
cd examples/cadzilla
altairsim drawdemo.toml
```

The machine boots itself to the CP/M prompt. Type **DRAWDEMO** :

```
56K CP/M 2.2b v2.3
For Altair 8" Floppy

A>DRAWDEMO

CADzilla drawdemo: the ACRTC drawing commands, 1024x768, 8 bpp

1/21 DOT - random dots placed by AMOVE; border dots stepped
        by RMOVE
Press any key for the next screen (ESC quits)...
```

On a build with **SDL3**, a 1024x768 window opens and comes to the front. The console tells you what each screen shows.

- Press any key for the next screen. A key typed in the window goes to CP/M, the same as a key typed in the terminal.
- Press **ESC** or **Ctrl-C** to stop drawdemo and go back to **A>**.
- Press **Ctrl-E** (STOP) to go to the monitor at any time. Type **RUN** to continue.

On a **headless** build, the machine runs the same program and draws nothing.

## Two speeds

| Machine file                    | Speed                                                                                                                                                                                                                     |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>drawdemo.toml</code>      | <b>Full speed.</b> The processor runs as fast as the host can run it. Each ACRTC command is complete as soon as it arrives, so each screen appears at once.                                                               |
| <code>drawdemo-real.toml</code> | <b>The speed of the real hardware.</b> A 2 MHz 8080 in real time, and each ACRTC command takes the time that the HD63484 data sheet gives. The filled shapes and the <code>PAINT</code> screen take visible time to draw. |

The two files use the same disk. `drawdemo-real.toml` starts from `drawdemo.toml` and changes two settings: `cpu0 clock_hz = 2000000` and `cad0 draw_rate = "real"`. You can also change them at the monitor with `SET cpu0 clock_hz=2000000` and `SET cad0 draw_rate=real`.

## The 21 screens

|    | Command            | What it draws                                                        |
|----|--------------------|----------------------------------------------------------------------|
| 1  | <code>DOT</code>   | Single pixels, placed with <code>AMOVE</code> and <code>RMOVE</code> |
| 2  | <code>ALINE</code> | Absolute lines                                                       |
| 3  | <code>RLINE</code> | Relative lines                                                       |
| 4  | <code>ARCT</code>  | Absolute rectangles                                                  |
| 5  | <code>RRCT</code>  | Relative rectangles                                                  |
| 6  | <code>APLL</code>  | Absolute polylines                                                   |
| 7  | <code>RPLL</code>  | Relative polylines                                                   |
| 8  | <code>APLG</code>  | Absolute polygons                                                    |
| 9  | <code>RPLG</code>  | Relative polygons                                                    |
| 10 | <code>AFRCT</code> | Absolute filled rectangles                                           |
| 11 | <code>RFRCT</code> | Relative filled rectangles                                           |
| 12 | <code>CRCL</code>  | Circles                                                              |
| 13 | <code>ELPS</code>  | Ellipses                                                             |
| 14 | <code>AARC</code>  | Circular arcs, absolute and relative ( <code>RARC</code> )           |
| 15 | <code>AEARC</code> | Elliptical arcs, absolute and relative ( <code>REARC</code> )        |
| 16 | <code>PAINT</code> | Area fill to an edge color, solid and with a pattern                 |
| 17 | <code>PTN</code>   | The pattern RAM, drawn as a picture                                  |
| 18 | <code>AGCPY</code> | Graphic copy, absolute and relative ( <code>RGCPY</code> )           |
| 19 |                    | The eight operation modes (OPM)                                      |
| 20 |                    | The color modes (COL), line styles and pattern zoom                  |
| 21 |                    | The area modes (AREA): clip, hole and stop                           |

## The machine

`drawdemo.toml` is the built-in `cadzilla` machine with the parts that CP/M needs added:

- An 8080 processor, an 88-2SIO for the console at port `10` , and 56K of RAM.
- The **CADzilla** board at ports `70` - `77` , set to a 1024x768 monitor. `drawdemo` sets the ACRTC for 1024x768, so the board's `mode` must agree.

- An 88-DCDD floppy controller at port `08` , with the disk in drive 0.
- The DBL boot PROM at `FF00` . `RUN FF00` is the whole startup, as on a real disk Altair.
- The host bridge at port `B0` , for `R` , `W` and `HDIR` .

## The disk

`cpm22b23-56k-drawdemo.dsk` is the CP/M 2.2b system disk from `examples/cpm` , changed to make room for drawdemo. That disk had 18K free, and `DRAWDEMO.COM` is 18K. These files were removed:

- `MBASIC.COM`
- the BASIC programs that need it: `LUNAR.BAS` , `STARINS.BAS` , `STARTRK.BAS` and `TICTAK.BAS`

The disk now has `DRAWDEMO.COM` and 60K free. All the other programs are still on it: `ASM` , `LOAD` , `DDT` , `M80` , `L80` , `ED` , `PIP` and the host-bridge utilities `R` , `W` and `HDIR` .

Drive 0 is read/write, as on a real machine. Nothing restores the disk after CP/M writes to it. Make a copy of the image before you test writes.

`DRAWDEMO.HEX` is the Intel HEX file of the program. CP/M's `LOAD` makes `DRAWDEMO.COM` from it.

`DRAWDEMO.ASM` is the source. `tools/make-drawdemo-disk.sh` makes the disk from the `examples/cpm` disk.

This example is in the source repository only. It is not in the release package.